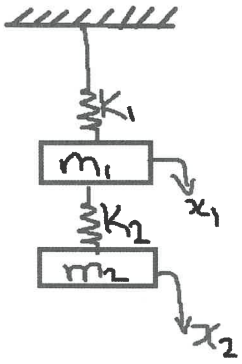
USN | | | | | | | | | | | [24MTEVT102]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

VEHICLE DYNAMICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Explain the significance of the roll-over, lift, and drag model in the design and safety evaluation of automobiles.	CO1	L2	**10**
	b.	Find the natural frequcny and mode shapes of the system shown in Fig.1b for $K_1=2k$, $K_2=K$ $m_1=2m$, $m_2=m$  Fig. 1b	CO1	L4	**10**
2.	**a.**	Classify the types of mechanical vibrations. Derive an expression for single degree of freedom system	CO2	L2	**10**
	b.	Derive an expression for logarithmic decrement	CO2	L3	**10**
3.	**a.**	Derive a differential equation for forced vibration by using graphical method.	CO3	L3	**10**
	b.	Explain simple harmonic motion with neat sketch	CO3	L2	**10**
		OR			

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4.	**a.**	Derive an expression for the weight distribution on front and rear vehicle of an engine	CO3	L3	**10**
	b.	Derive an expression for maximum tractive force.	CO3	L3	**10**
5.	**a.**	Explain the concepts of Roll Steer and Roll Centre in vehicle dynamics. How do they influence the handling characteristics of a vehicle?	CO4	L2	**10**
	b.	Elucidate the phenomena of wheel wobble and wheel shimmy in automotive systems	CO4	L3	**10**
		OR			
6.	**a.**	Describe roll axis and vehicle under the action of side forces	CO4	L3	**10**
	b.	Explain different parts in suspension system with a neat sketch	CO4	L2	**10**
7.	**a.**	Explain basics requirement of wheel also mention the basic constituents of a tyre.	CO5	L2	**10**
	b.	Illustrate the concepts of Oversteer and Understeer in vehicle dynamics. Discuss their causes.	CO5	L3	**10**
		OR			
8.	**a.**	Discuss the power consumed by a tire during vehicle operation.	CO5	L3	**10**
	b.	What is caster angle? explain its different type with a neat sketch.	CO5	L2	**10**